

SQL Quest!

The Super Fun Practice Book for Young Data Explorers

WELCOME

Hello, young explorer!

In this book you will learn to talk to computers using SQL, the language of data. Ready? Let’s go!



GETTING STARTED

Meet Your Adventure Crew!

Four friendly buddies will join you on every page of this quest.



Professor Hoot

Explains every new idea slowly and clearly.



Beep the Robot

Runs your SQL spells and shows the answers.



Dottie the Dragon

Loves challenges and cheers you on!



Tabby the Cat

Spots mistakes and shares handy tips.

GETTING STARTED

How to Use This Book



Every module has the same fun steps. Follow them and you will become an SQL Superstar!



1. Read

Read each page with a grown up or by yourself.



2. Look

Spells are on the left. Answers are on the right.



3. Try

On Your Turn pages, write your answer before peeking.



4. Win

Collect a badge at the end of every module!

GETTING STARTED

Your Quest Map



Start at 1 and hop all the way to 35. Colour in each stop when you earn its badge!

1. Data	2. Database	3. Tables	4. Hello SQL	5. Select All	6. Columns	7. Where
8. Compare	9. And Or	10. Not In	11. Between	12. Like	13. Order By	14. Limit
15. Distinct	16. Nicknames	17. Maths	18. Count	19. Sum Avg	20. Min Max	21. Group By
22. Having	23. Null	24. Create	25. Types	26. Insert	27. Update	28. Delete
29. Keys	30. Join	31. Left Join	32. Subquery	33. Case	34. Windows	35. With



Modules 1 to 4: Data basics. 5 to 17: Asking questions. 18 to 23: Counting and grouping. 24 to 29: Building tables. 30 to 35: Advanced adventures!

1

MODULE 1 OF 35

What Is Data?

Facts, facts everywhere! Let’s find out what data really is.

Mission: Spot data hiding all around you



MODULE 1 · DATA

Data Is Just Facts



Data means little facts about things. Your name, your age and your favourite colour are all data!



Words

Like names: Luna, Rex and Pip.



Numbers

Like ages and scores: 3, 7 or 100.



Yes or No

Like “Is it raining?” Yes!

BIG IDEA

Data is like stickers in a sticker book.

Each sticker shows one little fact. When we keep facts tidy, we can find them super fast!



MODULE 1 · DATA

Data Is Everywhere



Look around you! Data is hiding in lots of places.



Toy Box

How many toys do you have?



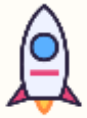
Games

Your high score is data!



School

The register lists who is here today.



Space

Scientists count stars and planets.

QUIZ · MODULE 1

Quick Quiz Time!



Which one is NOT a fact about a pet?

A Its name is Luna

B It is 5 years old

C Purple Tuesday jumps happily



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: C. That sentence is silly. It does not tell us a real fact!

YOUR TURN!

Be a Data Detective

Write down 3 facts about YOU: your name, your age and your favourite colour.

Write your answer here:

You can do it! Try first, then peek at the next page.



Hint: Each fact you write is one piece of data!

RECAP

Super Sticker Recap: What Is Data?



- Data means facts about things
- Data can be words, numbers or yes and no
- Data is everywhere around us



Data Spotter Badge Earned!

2

MODULE 2 OF 35

What Is a Database?

A super tidy toy box for facts!

Mission: Discover where computers keep their data



MODULE 2 · DATABASE

What Is a Database?



A database is a place where a computer keeps lots of data, all neat and tidy.



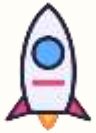
Like a Toy Box

Everything has its own special place.



Like a Library

Books sit on the right shelves.



Super Fast

The computer finds things in a blink!

BIG IDEA

A database holds TABLES, and tables hold data.

Think of a big cupboard with labelled drawers inside.



Databases Help Everyone



Lots of people use databases every single day.



Video Games

They remember your level and your coins.



Hospitals

Doctors keep notes safe and tidy.



Shops

Shops know which sweets are left.



Libraries

Find any book by its name.

MODULE 2 · DATABASE

Our Pet Party Database



In this book we explore a pretend database called **Pet Party**. It has two tables!

pets

8 pets with their names, ages, colours and treats.

owners

5 owners and the towns they live in.

Your Job

Ask clever questions to find the answers!

QUIZ · MODULE 2

Quick Quiz Time!



What does a database keep?

A Data, all neat and tidy

B Real puppies

C Rain clouds



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: A. A database keeps data, tidy and easy to find.

RECAP

Super Sticker Recap: What Is a Database?



- A database stores data neatly
- It keeps the data in tables
- Our database is called Pet Party



Database Explorer Badge Earned!

3

MODULE 3 OF 35

Tables, Rows and Columns

Meet the grids where data lives!

Mission: Learn the parts of a table



MODULE 3 · TABLES

A Table Is a Grid



A table looks like a grid, a bit like a chocolate bar with lots of squares!



Columns

Go up and down. Each column is one kind of fact.



Rows

Go across. Each row is one thing, like one pet.



Cells

Each little box holds one fact.

Spot the Row!



id	name	animal	age	colour	treats	owner_id
1	Biscuit	dog	3	brown	12	1
2	Luna	cat	5	black	7	2
3	Pip	rabbit	1	white	9	3
4	Rex	dog	7	gold	15	1
5	Mango	parrot	2	green	4	4
6	Coco	cat	4	white	10	2
7	Ziggy	hamster	1	orange	3	3
8	Bubbles	fish	2	orange	2	NULL

The yellow stripe is ONE row.

It tells us everything about Rex the dog.

Rows go ACROSS the table.

Spot the Column!



id	name	animal	age	colour	treats	owner_id
1	Biscuit	dog	3	brown	12	1
2	Luna	cat	5	black	7	2
3	Pip	rabbit	1	white	9	3
4	Rex	dog	7	gold	15	1
5	Mango	parrot	2	green	4	4
6	Coco	cat	4	white	10	2
7	Ziggy	hamster	1	orange	3	3
8	Bubbles	fish	2	orange	2	NULL

The pink words are ONE column.

This column is called **name**.

Columns go DOWN the table.

MODULE 3 · TABLES

Meet the Owners Table



id	name	town
1	Amara	Sunnyvale
2	Leo	Moonbridge
3	Priya	Sunnyvale
4	Kofi	Starfield
5	Mia	Moonbridge

This table has 5 rows: one for each owner.

It has 3 columns: id, name and town.

The id is each owner’s special number.

QUIZ · MODULE 3

Quick Quiz Time!



How many rows does the pets table have?

A 3

B 8

C 100



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: B. There are 8 pets, so there are 8 rows.

YOUR TURN!

Column Counter

Look back at the pets table. How many columns does it have? Write all their names.

You can do it! Try first, then peek at the next page.

Write your answer here:



Hint: Look at the very top row, the headings!

RECAP

Super Sticker Recap: Tables, Rows and Columns



- A table is a grid of data
- Rows go across, columns go down
- Each cell holds one fact



Table Master Badge Earned!

4

MODULE 4 OF 35

Hello, SQL!

The secret language for talking to databases

Mission: Learn what SQL is and how it works



MODULE 4 · HELLO SQL

What Is SQL?



SQL stands for **Structured Query Language**. It is how we ask a database questions!



A Magic Spell

You write words and the computer does the work.



A Query

A query is a question, like “Which pets are cats?”



Say It

Some say “S Q L” and some say “sequel”. Both are fine!

BIG IDEA

You ask. The database answers.

SQL is like talking to a super smart librarian robot.



MODULE 4 · HELLO SQL

Your Very First Spell



Here is the magic spell. Every piece has its own job!

SELECT	name	FROM	pets	;
show me	this column	from	this table	the end



Every SQL spell ends with a semicolon. It is a full stop for the computer.

MODULE 4 · HELLO SQL

Spell Rules



Follow these three rules and your spells will always work.



CAPITAL Keywords

We write words like SELECT in capitals so they are easy to spot.



Spelling Counts

Write pets, not pots! The computer is very fussy.



Semicolon

Finish with ; so the computer knows you are done.

QUIZ · MODULE 4

Quick Quiz Time!



What does SQL help us do?

A Bake a cake

B Ask a database questions

C Paint a picture



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: B. SQL is the language we use to ask databases questions.

RECAP

Super Sticker Recap: Hello, SQL!



- SQL is a language for databases
- A query is a question
- Spells end with a semicolon



SQL Starter Badge Earned!

5

MODULE 5 OF 35

SELECT Everything!

One tiny star shows the whole table

Mission: Use SELECT * to see all the data



MODULE 5 · SELECT ALL

The Star Means Everything



In SQL, the star * means ALL the columns. So SELECT * shows everything!



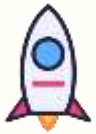
* Means All

Every single column comes back.



Great for Peeking

Use it to see what a table holds.



Super Short

Just three words and a star!

MODULE 5 · SELECT ALL

The Show Everything Spell



Here is the magic spell. Every piece has its own job!

SELECT	*	FROM	pets	;
show me	every column	from	the pets table	the end



Read it like this: show me every column from the pets table.

MODULE 5 · SELECT ALL

Show Every Pet



YOUR SQL SPELL

```
SELECT * FROM pets;
```

Wow! All 8 pets and all 7 columns came back. That is the whole table!

RUN

WHAT THE COMPUTER SHOWS

id	name	animal	age	colour	treats	owner_id
1	Biscuit	dog	3	brown	12	1
2	Luna	cat	5	black	7	2
3	Pip	rabbit	1	white	9	3
4	Rex	dog	7	gold	15	1
5	Mango	parrot	2	green	4	4
6	Coco	cat	4	white	10	2
7	Ziggy	hamster	1	orange	3	3
8	Bubbles	fish	2	orange	2	NULL

8 rows came back!

YOUR TURN!

Peek at the Owners



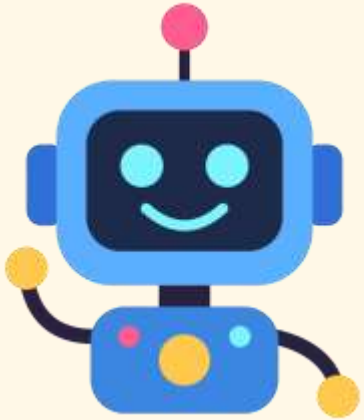
Write a spell to show EVERYTHING in the owners table.

FILL IN THE BLANKS

SELECT ___ FROM _____;

Write your answer here:

You can do it! Try first, then peek at the next page.



Hint: Use the star, then the table name.

ANSWER CHECK

Answer: All the Owners



YOUR SQL SPELL

```
SELECT * FROM owners;
```

Brilliant! 5 owners with their id, name and town.

RUN

WHAT THE COMPUTER SHOWS

id	name	town
1	Amara	Sunnyvale
2	Leo	Moonbridge
3	Priya	Sunnyvale
4	Kofi	Starfield
5	Mia	Moonbridge

5 rows came back!

RECAP

Super Sticker Recap: SELECT Everything!



- * means all the columns
- SELECT * FROM a table shows everything
- It is great for peeking at a table



Star Seeker Badge Earned!

6

MODULE 6 OF 35

Pick Your Columns

Choose only the facts you want

Mission: Select one or more columns by name



MODULE 6 · COLUMNS

Why Pick Columns?



Sometimes we only want a few facts, not the whole table. So we name the columns we want!



Less Clutter

See only what you need.



Use Commas

Put a comma between column names.



Your Order

Columns come back in the order you write them.

The Pick Columns Spell



Here is the magic spell. Every piece has its own job!

SELECT	name, animal	FROM	pets	;
show me	these two columns	from	the pets table	the end



The comma is like saying “and”. Show me name AND animal.

MODULE 6 · COLUMNS

Names and Animals



YOUR SQL SPELL

```
SELECT name, animal
FROM pets;
```



Only 2 columns came back: name and animal. So neat!

RUN

WHAT THE COMPUTER SHOWS

name	animal
Biscuit	dog
Luna	cat
Pip	rabbit
Rex	dog
Mango	parrot
Coco	cat
Ziggy	hamster
Bubbles	fish

8 rows came back!

MODULE 6 · COLUMNS

Who Ate the Treats?



YOUR SQL SPELL

```
SELECT name, treats
FROM pets;
```

Now we see each pet's name with how many treats it ate.

RUN

WHAT THE COMPUTER SHOWS

name	treats
Biscuit	12
Luna	7
Pip	9
Rex	15
Mango	4
Coco	10
Ziggy	3
Bubbles	2

8 rows came back!

Just One Column



YOUR SQL SPELL

```
SELECT town
FROM owners;
```



One column is fine too! Here are all the towns.

RUN

WHAT THE COMPUTER SHOWS

town
Sunnyvale
Moonbridge
Sunnyvale
Starfield
Moonbridge

5 rows came back!

MODULE 6 · COLUMNS

Comma Care



Oops!

```sql
SELECT name, animal,
FROM pets;
```

Hooray!

```sql
SELECT name, animal
FROM pets;
```

No comma after the LAST column! A comma there confuses the computer.

YOUR TURN!

Names and Colours

Show the name and colour of every pet.

You can do it! Try first, then peek at the next page.

FILL IN THE BLANKS

```
SELECT ____, _____
FROM pets;
```



Write your answer here:

Hint: Remember the comma between the two columns.

ANSWER CHECK

Answer: Names and Colours



YOUR SQL SPELL

```
SELECT name, colour
FROM pets;
```

Super! Each pet's name sits next to its colour.

RUN

WHAT THE COMPUTER SHOWS

name	colour
Biscuit	brown
Luna	black
Pip	white
Rex	gold
Mango	green
Coco	white
Ziggy	orange
Bubbles	orange

8 rows came back!

RECAP

Super Sticker Recap: Pick Your Columns



- Name the columns you want
- Put commas between them
- No comma after the last one



Column Picker Badge Earned!

7

MODULE 7 OF 35

WHERE: The Super Filter

Find exactly the rows you want

Mission: Use WHERE to filter rows



WHERE Is a Sieve



WHERE works like a sieve. It lets through only the rows that match your rule!



A Rule

WHERE animal = 'dog' means dogs only.



Match: Keep

Rows that match are shown.



No Match: Hide

All the other rows stay hidden.

MODULE 7 · WHERE

The Filter Spell



Here is the magic spell. Every piece has its own job!

```
SELECT *  FROM pets  WHERE  animal = 'dog'  ;
```

show everything — **from pets** — **but only where** — **the animal is dog** — **the end**



WHERE always comes AFTER the FROM part.

MODULE 7 · WHERE

Only the Dogs



YOUR SQL SPELL

```
SELECT name, animal
FROM pets
WHERE animal = 'dog';
```

RUN

WHAT THE COMPUTER SHOWS

name	animal
Biscuit	dog
Rex	dog

2 rows came back!

Only Biscuit and Rex are dogs, so only they came back!

MODULE 7 · WHERE

White Pets Only



YOUR SQL SPELL

```
SELECT name, colour
FROM pets
WHERE colour = 'white';
```

RUN

WHAT THE COMPUTER SHOWS

name	colour
Pip	white
Coco	white

2 rows came back!

Pip and Coco are white. The sieve kept just those two.

MODULE 7 · WHERE

Owners in Sunnyvale



YOUR SQL SPELL

```
SELECT name, town
FROM owners
WHERE town = 'Sunnyvale';
```

RUN

WHAT THE COMPUTER SHOWS

name	town
Amara	Sunnyvale
Priya	Sunnyvale

2 rows came back!

Amara and Priya both live in Sunnyvale.

Quote Marks for Words



Oops!

```sql
SELECT name
FROM pets
WHERE animal = dog;
```

Hooray!

```sql
SELECT name
FROM pets
WHERE animal = 'dog';
```



Words need single quote marks around them. Numbers do not!

YOUR TURN!

Find the Cats



Show the names of all the cats.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name
FROM pets
WHERE _____ = '___';
```

Write your answer here:

Hint: The column is animal and the word is cat.

ANSWER CHECK

Answer: The Cats



YOUR SQL SPELL

```
SELECT name
FROM pets
WHERE animal = 'cat';
```

Luna and Coco! Two purring cats.

RUN

WHAT THE COMPUTER SHOWS

name
Luna
Coco

2 rows came back!

RECAP

Super Sticker Recap: WHERE: The Super Filter



- WHERE keeps only matching rows
- WHERE comes after FROM
- Words go inside single quotes



Filter Hero Badge Earned!

8

MODULE 8 OF 35

Comparison Signs

Bigger, smaller, the same or different?

Mission: Compare numbers and words in WHERE



MODULE 8 · COMPARE

Meet the Signs



These signs help us compare things inside WHERE.

= **Equal to** age = 3 means exactly 3	> **Bigger than** age > 3 means 4, 5, 6...	< **Smaller than** age < 3 means 2, 1, 0
>= **Bigger or equal** age >= 3 includes 3	<= **Smaller or equal** age <= 3 includes 3	<> **Not equal** age <> 3 means not 3

BIG IDEA

The hungry crocodile always eats the BIGGER number!

The open mouth of > and < always faces the bigger side.



MODULE 8 · COMPARE

Older Than 3



YOUR SQL SPELL

```
SELECT name, age
FROM pets
WHERE age > 3;
```

RUN

WHAT THE COMPUTER SHOWS

name	age
Luna	5
Rex	7
Coco	4

3 rows came back!

Luna, Rex and Coco are older than 3. Age 3 itself is NOT bigger than 3!

MODULE 8 · COMPARE

Small Snackers



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
WHERE treats <= 4;
```

Mango, Ziggy and Bubbles ate 4 treats or fewer.

RUN

WHAT THE COMPUTER SHOWS

name	treats
Mango	4
Ziggy	3
Bubbles	2

3 rows came back!

MODULE 8 · COMPARE

Everyone But Dogs



YOUR SQL SPELL

```
SELECT name, animal
FROM pets
WHERE animal <> 'dog';
```

The <> sign means not equal. All 6 pets that are not dogs came back!

RUN

WHAT THE COMPUTER SHOWS

name	animal
Luna	cat
Pip	rabbit
Mango	parrot
Coco	cat
Ziggy	hamster
Bubbles	fish

6 rows came back!

QUIZ · MODULE 8

Quick Quiz Time!



Which spell finds pets aged 5 or more?

A WHERE age > 5

B WHERE age >= 5

C WHERE age < 5



Choose A, B or C. Then turn the page upside down to check the yellow box!

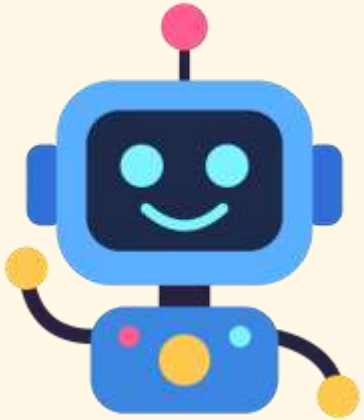
Answer: B. The >= sign includes 5 itself.

YOUR TURN!

Big Eaters

Show the name and treats of pets that ate MORE than 9 treats.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, treats
FROM pets
WHERE treats ___ ___;
```

Write your answer here:

Hint: Bigger than uses the > sign.

ANSWER CHECK

Answer: Big Eaters



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
WHERE treats > 9;
```

RUN

Biscuit, Rex and Coco are the big eaters!

WHAT THE COMPUTER SHOWS

name	treats
Biscuit	12
Rex	15
Coco	10

3 rows came back!

RECAP

Super Sticker Recap: Comparison Signs



- = means the same, <> means different
- > is bigger and < is smaller
- >= and <= include the number itself



Sign Wizard Badge Earned!

9

MODULE 9 OF 35

AND and OR

Join two rules together

Mission: Combine rules with AND and OR



MODULE 9 · AND OR

Two Rules at Once



AND and OR let us use two rules inside one WHERE.



AND

BOTH rules must be true. Very picky!



OR

At least ONE rule must be true. Very friendly!



Example

A white cat? Use AND. A cat or a dog? Use OR.

BIG IDEA

AND is picky. OR is friendly.

AND wants everything to be true. OR is happy with just one.



MODULE 9 · AND OR

White Cats Only



YOUR SQL SPELL

```
SELECT name, animal, colour
FROM pets
WHERE animal = 'cat'
AND colour = 'white';
```

RUN

WHAT THE COMPUTER SHOWS

name	animal	colour
Coco	cat	white

1 row came back!

Only Coco is a cat AND white. Luna is black, so she stays hidden.

Parrots or Fish



YOUR SQL SPELL

```
SELECT name, animal
FROM pets
WHERE animal = 'parrot'
OR animal = 'fish';
```

RUN

WHAT THE COMPUTER SHOWS

name	animal
Mango	parrot
Bubbles	fish

2 rows came back!

Mango is a parrot and Bubbles is a fish. Either one is fine with OR!

Young and Hungry



YOUR SQL SPELL

```
SELECT name, age, treats
FROM pets
WHERE age < 3
AND treats > 5;
```

RUN

WHAT THE COMPUTER SHOWS

name	age	treats
Pip	1	9

1 row came back!

Only Pip is younger than 3 AND ate more than 5 treats.

YOUR TURN!

Super Dogs

Show the dogs that ate more than 12 treats.

You can do it! Try first, then peek at the next page.

FILL IN THE BLANKS

```
SELECT name, treats
FROM pets
WHERE animal = 'dog'
___ treats > 12;
```



Hint: Both rules must be true.

ANSWER CHECK

Answer: Super Dogs



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
WHERE animal = 'dog'
AND treats > 12;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats
Rex	15

1 row came back!

Rex! He is a dog AND he ate 15 treats.

RECAP

Super Sticker Recap: AND and OR



- AND needs both rules to be true
- OR needs just one rule to be true
- You can join many rules together



Rule Combiner Badge Earned!

10

MODULE 10 OF 35

NOT and IN

Shortcuts for picky questions

Mission: Use IN for lists and NOT to flip a rule



MODULE 10 · NOT IN

Two Handy Helpers



IN checks a list. NOT flips a rule upside down!



IN

Is it on my list? Like a party guest list.



NOT

Turns yes into no, and no into yes.



NOT IN

Anything that is NOT on the list.

MODULE 10 · NOT IN

Cats and Dogs



YOUR SQL SPELL

```
SELECT name, animal
FROM pets
WHERE animal IN ('cat', 'dog');
```

IN is a shortcut for lots of ORs. Cats and dogs only!

RUN

WHAT THE COMPUTER SHOWS

name	animal
Biscuit	dog
Luna	cat
Rex	dog
Coco	cat

4 rows came back!

MODULE 10 · NOT IN

No Cats, No Dogs



YOUR SQL SPELL

```
SELECT name, animal
FROM pets
WHERE animal NOT IN ('cat', 'dog');
```

Flip it! Every pet that is not a cat or a dog.

RUN

WHAT THE COMPUTER SHOWS

name	animal
Pip	rabbit
Mango	parrot
Ziggy	hamster
Bubbles	fish

4 rows came back!

MODULE 10 · NOT IN

Not Orange



YOUR SQL SPELL

```
SELECT name, colour
FROM pets
WHERE NOT colour = 'orange';
```

NOT hides the orange pets, Ziggy and Bubbles.

RUN

WHAT THE COMPUTER SHOWS

name	colour
Biscuit	brown
Luna	black
Pip	white
Rex	gold
Mango	green
Coco	white

6 rows came back!

QUIZ · MODULE 10

Quick Quiz Time!



Which spell finds pets that are a rabbit OR a hamster?

A WHERE animal IN ('rabbit', 'hamster')

B WHERE animal NOT IN ('rabbit', 'hamster')

C WHERE NOT animal = 'rabbit'



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: A. IN checks the list: rabbit or hamster.

YOUR TURN!

Party Guests

Show the pets called Pip, Rex or Mango using IN.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, animal
FROM pets
WHERE name IN (_____, _____, _____);
```

Write your answer here:

Hint: Put each name in single quotes, with commas between.

ANSWER CHECK

Answer: Party Guests



YOUR SQL SPELL

```
SELECT name, animal
FROM pets
WHERE name IN ('Pip', 'Rex', 'Mango');
```

All three party guests have arrived!

RUN

WHAT THE COMPUTER SHOWS

name	animal
Pip	rabbit
Rex	dog
Mango	parrot

3 rows came back!

RECAP

Super Sticker Recap: NOT and IN



- IN checks if something is on a list
- NOT flips a rule
- NOT IN finds things that are not on the list



List Checker Badge Earned!

11

MODULE 11 OF 35

BETWEEN

Find numbers inside a fence

Mission: Use BETWEEN for ranges



MODULE 11 · BETWEEN

A Number Fence



BETWEEN finds numbers from one number up to another. Both ends count!



Two Ends

BETWEEN 2 AND 4 means 2, 3 or 4.



Ends Included

2 and 4 are inside the fence too.



Same As

age >= 2 AND age <= 4

Ages 2 to 4



YOUR SQL SPELL

```
SELECT name, age
FROM pets
WHERE age BETWEEN 2 AND 4;
```

Biscuit, Mango, Coco and Bubbles are all aged 2, 3 or 4.

RUN

WHAT THE COMPUTER SHOWS

name	age
Biscuit	3
Mango	2
Coco	4
Bubbles	2

4 rows came back!

MODULE 11 · BETWEEN

Middle Snackers



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
WHERE treats BETWEEN 5 AND 10;
```

Luna, Pip and Coco ate between 5 and 10 treats.

RUN

WHAT THE COMPUTER SHOWS

name	treats
Luna	7
Pip	9
Coco	10

3 rows came back!

QUIZ · MODULE 11

Quick Quiz Time!



Is age 4 inside BETWEEN 2 AND 4?

A Yes, the ends count!

B No, never

C Only on Mondays



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: A. BETWEEN includes both end numbers.

YOUR TURN!

Grown Up Pets

Show the pets aged between 4 and 7.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, age
FROM pets
WHERE age _____ 4 ___ 7;
```

Write your answer here:

Hint: BETWEEN a number AND a number.

ANSWER CHECK

Answer: Grown Up Pets



YOUR SQL SPELL

```
SELECT name, age
FROM pets
WHERE age BETWEEN 4 AND 7;
```

Luna, Rex and Coco are inside the fence.

RUN

WHAT THE COMPUTER SHOWS

name	age
Luna	5
Rex	7
Coco	4

3 rows came back!

RECAP

Super Sticker Recap: BETWEEN



- BETWEEN finds a range of values
- Both end numbers are included
- Write it as BETWEEN low AND high



Fence Builder Badge Earned!

12

MODULE 12 OF 35

LIKE and Wildcards

Search with secret jokers

Mission: Find words that match a pattern



MODULE 12 · LIKE

Wildcard Jokers



LIKE searches for patterns in words. Wildcards are jokers that stand for missing letters!



The % Joker

Stands for ANY letters, or even none at all.



The _ Joker

Stands for exactly ONE letter.



Example

'B%' means “starts with B”.

MODULE 12 · LIKE

Starts With B



YOUR SQL SPELL

```
SELECT name
FROM pets
WHERE name LIKE 'B%';
```

RUN

WHAT THE COMPUTER SHOWS

name
Biscuit
Bubbles

2 rows came back!

Biscuit and Bubbles both start with B!

MODULE 12 · LIKE

Ends With O



YOUR SQL SPELL

```
SELECT name
FROM pets
WHERE name LIKE '%o';
```

Mango and Coco both end with the letter o.

RUN

WHAT THE COMPUTER SHOWS

name
Mango
Coco

2 rows came back!

MODULE 12 · LIKE

Second Letter E



YOUR SQL SPELL

```
SELECT name
FROM pets
WHERE name LIKE '_e%';
```

RUN

WHAT THE COMPUTER SHOWS

name
Rex

1 row came back!

One joker letter, then e, then anything. That is Rex!

YOUR TURN!

The Hidden I

Find every pet with the letter i anywhere in its name.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name
FROM pets
WHERE name LIKE '____';
```

Write your answer here:

Hint: Put a % joker on BOTH sides of the i.

ANSWER CHECK

Answer: The Hidden I



YOUR SQL SPELL

```
SELECT name
FROM pets
WHERE name LIKE '%i%';
```

Biscuit, Pip and Ziggy all have an i hiding inside.

RUN

WHAT THE COMPUTER SHOWS

name
Biscuit
Pip
Ziggy

3 rows came back!

RECAP

Super Sticker Recap: LIKE and Wildcards



- LIKE searches for patterns
- % means any letters
- _ means exactly one letter



Pattern Finder Badge Earned!

13

MODULE 13 OF 35

ORDER BY

Line everything up nicely

Mission: Sort rows from small to big, or big to small



MODULE 13 · ORDER BY

Lining Up



ORDER BY sorts the rows, just like lining up by height in class!



ASC

Small to big, A to Z. This is the normal way.



DESC

Big to small, Z to A. Upside down!



Near the End

ORDER BY goes near the end of the spell.

MODULE 13 · ORDER BY

Youngest First



YOUR SQL SPELL

```
SELECT name, age
FROM pets
ORDER BY age;
```

From the babies aged 1, all the way up to Rex aged 7.

RUN

WHAT THE COMPUTER SHOWS

name	age
Pip	1
Ziggy	1
Mango	2
Bubbles	2
Biscuit	3
Coco	4
Luna	5
Rex	7

8 rows came back!

MODULE 13 · ORDER BY

Biggest Eaters First



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
ORDER BY treats DESC;
```

DESC flips it. Rex ate the most, so he is first!

RUN

WHAT THE COMPUTER SHOWS

name	treats
Rex	15
Biscuit	12
Coco	10
Pip	9
Luna	7
Mango	4
Ziggy	3
Bubbles	2

8 rows came back!

MODULE 13 · ORDER BY

A to Z



YOUR SQL SPELL

```
SELECT name
FROM owners
ORDER BY name;
```

Words sort in alphabet order: Amara, Kofi, Leo, Mia, Priya.

RUN

WHAT THE COMPUTER SHOWS

name
Amara
Kofi
Leo
Mia
Priya

5 rows came back!

YOUR TURN!

Oldest First



Show pet names and ages, with the OLDEST pet at the top.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, age
FROM pets
ORDER BY ___ ____;
```

Write your answer here:

Hint: Big to small uses DESC.

ANSWER CHECK

Answer: Oldest First



YOUR SQL SPELL

```
SELECT name, age
FROM pets
ORDER BY age DESC;
```

Rex is the oldest at 7 years old!

RUN

WHAT THE COMPUTER SHOWS

name	age
Rex	7
Luna	5
Coco	4
Biscuit	3
Mango	2
Bubbles	2
Pip	1
Ziggy	1

8 rows came back!

RECAP

Super Sticker Recap: ORDER BY



- ORDER BY sorts the rows
- ASC is small to big
- DESC is big to small



Line Leader Badge Earned!

14

MODULE 14 OF 35

LIMIT

Only show the first few

Mission: Stop after a set number of rows



MODULE 14 · LIMIT

Just a Few, Please



LIMIT tells the computer how many rows to show. Like taking just 3 sweets from the jar!



LIMIT 3

Only 3 rows come back.



Top Lists

Use it with ORDER BY for top 3 lists.



Very Last

LIMIT goes at the very end of the spell.

MODULE 14 · LIMIT

Top 3 Snackers



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
ORDER BY treats DESC
LIMIT 3;
```

The gold, silver and bronze snack champions!

RUN

WHAT THE COMPUTER SHOWS

name	treats
Rex	15
Biscuit	12
Coco	10

3 rows came back!

Just Two Owners



YOUR SQL SPELL

```
SELECT name
FROM owners
LIMIT 2;
```

RUN

WHAT THE COMPUTER SHOWS

name
Amara
Leo

2 rows came back!



We asked for 2 rows, so we got exactly 2 rows.

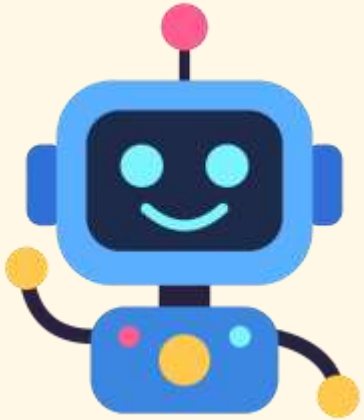
YOUR TURN!

Two Youngest



Show the 2 youngest pets.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, age
FROM pets
ORDER BY age
_____ _;
```

Hint: LIMIT and then a number.

ANSWER CHECK

Answer: Two Youngest



YOUR SQL SPELL

```
SELECT name, age
FROM pets
ORDER BY age
LIMIT 2;
```

RUN

WHAT THE COMPUTER SHOWS

name	age
Pip	1
Ziggy	1

2 rows came back!

Pip and Ziggy are both only 1 year old!

RECAP

Super Sticker Recap: LIMIT



- LIMIT shows only some rows
- LIMIT goes at the very end
- ORDER BY plus LIMIT makes top lists



Top Picker Badge Earned!

15

MODULE 15 OF 35

DISTINCT

No copycats allowed!

Mission: Remove repeated values



MODULE 15 · DISTINCT

Goodbye, Copies!



DISTINCT removes repeats, so each value shows up only once.



Unique

Each value appears just once.



Great For

Finding out what kinds of things exist.



Easy

Just add DISTINCT right after SELECT.

MODULE 15 · DISTINCT

Animals With Repeats



YOUR SQL SPELL

```
SELECT animal
FROM pets;
```



See? dog and cat show up twice each. Copycats!

RUN

WHAT THE COMPUTER SHOWS

animal
dog
cat
rabbit
dog
parrot
cat
hamster
fish

8 rows came back!

MODULE 15 · DISTINCT

Animals, No Repeats



YOUR SQL SPELL

```
SELECT DISTINCT animal
FROM pets;
```

Now each animal shows up only once. 6 kinds of animal!

RUN

WHAT THE COMPUTER SHOWS

animal
dog
cat
rabbit
parrot
hamster
fish

6 rows came back!

YOUR TURN!

Which Colours?

Show each pet colour only once.

FILL IN THE BLANKS

```
SELECT _____ colour
FROM pets;
```

Write your answer here:

You can do it! Try first, then peek at the next page.



Hint: This word removes the copies.

ANSWER CHECK

Answer: Which Colours?



YOUR SQL SPELL

```
SELECT DISTINCT colour
FROM pets;
```

6 different colours, and no repeats.

RUN

WHAT THE COMPUTER SHOWS

colour
brown
black
white
gold
green
orange

6 rows came back!

RECAP

Super Sticker Recap: DISTINCT



- DISTINCT removes repeats
- It goes right after SELECT
- It shows what kinds of values exist



Copycat Catcher Badge Earned!

16

MODULE 16 OF 35

AS: Nicknames

Give your columns fun new names

Mission: Rename columns in the answer



MODULE 16 · NICKNAMES

Column Nicknames



AS gives a column a nickname in the answer. The real table does not change!



Friendly Names

Make headings easy to read.



Only Pretend

The real column keeps its name.



The Pattern

column AS nickname

MODULE 16 · NICKNAMES

Pet Names



YOUR SQL SPELL

```
SELECT name AS pet_name
FROM pets
LIMIT 4;
```



The heading now says pet_name instead of name.

RUN

WHAT THE COMPUTER SHOWS

pet_name
Biscuit
Luna
Pip
Rex

4 rows came back!

MODULE 16 · NICKNAMES

Snack Counter



YOUR SQL SPELL

```
SELECT name AS buddy,
treats AS snacks
FROM pets
LIMIT 4;
```

RUN

WHAT THE COMPUTER SHOWS

buddy	snacks
Biscuit	12
Luna	7
Pip	9
Rex	15

4 rows came back!

Two nicknames at once: buddy and snacks!

YOUR TURN!

Home Towns



Show owners' names and towns, with the town heading called home.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, town __ _____
FROM owners;
```

Write your answer here:

Hint: column AS nickname

ANSWER CHECK

Answer: Home Towns



YOUR SQL SPELL

```
SELECT name, town AS home
FROM owners;
```

The town column now wears the nickname home.

RUN

WHAT THE COMPUTER SHOWS

name	home
Amara	Sunnyvale
Leo	Moonbridge
Priya	Sunnyvale
Kofi	Starfield
Mia	Moonbridge

5 rows came back!

RECAP

Super Sticker Recap: AS: Nicknames



- AS renames a column in the answer
- The real table does not change
- Nicknames make results easy to read



Nickname Maker Badge Earned!

17

MODULE 17 OF 35

Maths Magic

Make the computer do your sums

Mission: Use plus, times and divide inside SELECT



MODULE 17 · MATHS

The Computer Can Count!



SQL can do maths for every row in a flash.



+ Adds

treats + 5 gives 5 more.



* Times

treats * 2 means double!



/ Shares

treats / 2 shares in half.

MODULE 17 · MATHS

Double Treats



YOUR SQL SPELL

```
SELECT name, treats,
treats * 2 AS doubled
FROM pets
LIMIT 5;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats	doubled
Biscuit	12	24
Luna	7	14
Pip	9	18
Rex	15	30
Mango	4	8

5 rows came back!

Every pet’s treats, doubled. Yum!

MODULE 17 · MATHS

Times 7 Ages



YOUR SQL SPELL

```
SELECT name, age,
age * 7 AS times_seven
FROM pets
LIMIT 5;
```

RUN

WHAT THE COMPUTER SHOWS

name	age	times_seven
Biscuit	3	21
Luna	5	35
Pip	1	7
Rex	7	49
Mango	2	14

5 rows came back!

Just for fun, we times each age by 7. Rex gets 49!

MODULE 17 · MATHS

Birthday Bonus



YOUR SQL SPELL

```
SELECT name,
treats + 5 AS party_treats
FROM pets
LIMIT 5;
```

RUN

WHAT THE COMPUTER SHOWS

name	party_treats
Biscuit	17
Luna	12
Pip	14
Rex	20
Mango	9

5 rows came back!



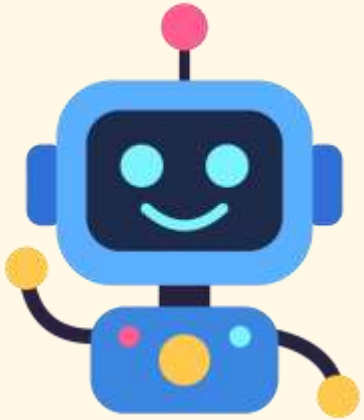
It is a party! Everyone gets 5 extra treats.

YOUR TURN!

Half Shares

Show each pet's name and its treats shared in half, called half.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name, treats _ _ AS half
FROM pets;
```

Write your answer here:

Hint: Sharing in half means divide by 2.

ANSWER CHECK

Answer: Half Shares



YOUR SQL SPELL

```
SELECT name, treats / 2.0 AS half
FROM pets;
```

We wrote 2.0 so the computer keeps the halves, like 3.5!

RUN

WHAT THE COMPUTER SHOWS

name	half
Biscuit	6.0
Luna	3.5
Pip	4.5
Rex	7.5
Mango	2.0
Coco	5.0
Ziggy	1.5
Bubbles	1.0

8 rows came back!

RECAP

Super Sticker Recap: Maths Magic



- + adds and * multiplies
- / divides: use 2.0 to keep the halves
- Give maths columns a nickname with AS



Maths Magician Badge Earned!

18

MODULE 18 OF 35

COUNT

How many are there?

Mission: Count rows with COUNT



MODULE 18 · COUNT

Counting Made Easy



COUNT(*) counts the rows for you. No fingers needed!



COUNT(*)

Counts every row.



With WHERE

Count only the rows that match.



One Answer

You get back one number.

MODULE 18 · COUNT

How Many Pets?



YOUR SQL SPELL

```
SELECT COUNT(*) AS total
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

total
8

1 row came back!

The computer counted 8 pets.

How Many Cats?



YOUR SQL SPELL

```
SELECT COUNT(*) AS cats
FROM pets
WHERE animal = 'cat';
```

RUN

WHAT THE COMPUTER SHOWS

cats
2

1 row came back!

The sieve keeps the cats, then COUNT counts them: 2!

YOUR TURN!

Count the Owners

How many owners live in Moonbridge?

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT _____(*)
FROM owners
WHERE town = '_____';
```

Write your answer here:

Hint: Use COUNT and the town name in quotes.

ANSWER CHECK

Answer: Moonbridge Owners



YOUR SQL SPELL

```
SELECT COUNT(*) AS owners
FROM owners
WHERE town = 'Moonbridge';
```

RUN

WHAT THE COMPUTER SHOWS

owners
2

1 row came back!

Leo and Mia live there, so the answer is 2.

RECAP

Super Sticker Recap: COUNT



- COUNT(*) counts rows
- Add WHERE to count only some
- COUNT gives back one number



Counting Champ Badge Earned!

19

MODULE 19 OF 35

SUM and AVG

Add it all up, then share it fairly

Mission: Find totals and averages



Two Number Helpers



SUM adds up a column. AVG finds the fair share, called the average.



SUM

Adds every number together.



AVG

If we shared fairly, how much would each get?



Numbers Only

They work on number columns.

BIG IDEA

8 pets ate 62 treats.

Shared out fairly, that is 7.75 treats each. That fair share is called the AVERAGE!



All the Treats



YOUR SQL SPELL

```
SELECT SUM(treats) AS all_treats
FROM pets;
```



12 + 7 + 9 + 15 + 4 + 10 + 3 + 2 makes 62 treats!

RUN

WHAT THE COMPUTER SHOWS

all_treats
62

1 row came back!

MODULE 19 · SUM AVG

Fair Share



YOUR SQL SPELL

```
SELECT AVG(treats) AS fair_share
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

fair_share
7.75

1 row came back!

62 treats shared by 8 pets is 7.75 treats each.

MODULE 19 · SUM AVG

Round It



YOUR SQL SPELL

```
SELECT ROUND(AVG(age), 1) AS avg_age
FROM pets;
```

🔍 ROUND tidies long numbers. The average age is about 3.1 years.

RUN

WHAT THE COMPUTER SHOWS

avg_age
3.1

1 row came back!

YOUR TURN!

Dog Treat Total

Add up all the treats eaten by dogs.

FILL IN THE BLANKS

```
SELECT ___(treats)
FROM pets
WHERE animal = '___';
```

Write your answer here:

You can do it! Try first, then peek at the next page.



Hint: SUM adds things up.

ANSWER CHECK

Answer: Dog Treat Total



YOUR SQL SPELL

```
SELECT SUM(treats) AS dog_treats
FROM pets
WHERE animal = 'dog';
```

RUN

WHAT THE COMPUTER SHOWS

dog_treats
27

1 row came back!

Biscuit ate 12 and Rex ate 15. That makes 27!

RECAP

Super Sticker Recap: SUM and AVG



- SUM adds a column up
- AVG finds the fair share
- ROUND tidies long numbers



Number Cruncher Badge Earned!

20

MODULE 20 OF 35

MIN and MAX

Find the smallest and the biggest

Mission: Discover the lowest and highest values



MODULE 20 · MIN MAX

Tiny and Giant



MIN finds the smallest value. MAX finds the biggest value.



MIN

The smallest, like the baby of the group.



MAX

The biggest, like the champion!



Words Too

MIN on names gives the first in ABC order.

Youngest and Oldest



YOUR SQL SPELL

```
SELECT MIN(age) AS youngest,
MAX(age) AS oldest
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

youngest	oldest
1	7

1 row came back!

The youngest pet is 1 and the oldest is 7.

Snack Record



YOUR SQL SPELL

```
SELECT MAX(treats) AS record
FROM pets;
```



The treat record is 15. Who holds it? Rex!

RUN

WHAT THE COMPUTER SHOWS

record
15

1 row came back!

MODULE 20 · MIN MAX

First in ABC



YOUR SQL SPELL

```
SELECT MIN(name) AS first_name
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

first_name
Biscuit

1 row came back!

Biscuit comes first in alphabet order.

YOUR TURN!

Cat Champion

What is the MOST treats eaten by a cat?

FILL IN THE BLANKS

```
SELECT ___(treats)
FROM pets
WHERE animal = 'cat';
```

Write your answer here:

You can do it! Try first, then peek at the next page.



Hint: Biggest means MAX.

ANSWER CHECK

Answer: Cat Champion



YOUR SQL SPELL

```
SELECT MAX(treats) AS cat_record
FROM pets
WHERE animal = 'cat';
```

RUN

WHAT THE COMPUTER SHOWS

cat_record
10

1 row came back!

Coco ate 10 treats, beating Luna’s 7.

RECAP

Super Sticker Recap: MIN and MAX



- MIN finds the smallest
- MAX finds the biggest
- They work on words too



Record Breaker Badge Earned!

21

MODULE 21 OF 35

GROUP BY

Sort rows into buckets

Mission: Group rows and count each bucket



MODULE 21 · GROUP BY

Buckets of Data



GROUP BY puts rows into buckets. Then COUNT or SUM works on each bucket!



Make Buckets

All dogs in one bucket, all cats in another.



Count Each

COUNT tells how many are in each bucket.



One Row Each

Each bucket gives back one row.

Our Animal Buckets



cat
Luna and Coco

2 pets

dog
Biscuit and Rex

2 pets

fish
Bubbles

1 pet

hamster
Ziggy

1 pet

parrot
Mango

1 pet

rabbit
Pip

1 pet

6 kinds of animal means 6 buckets. GROUP BY animal makes exactly these!

Pets per Animal



YOUR SQL SPELL

```
SELECT animal, COUNT(*) AS how_many
FROM pets
GROUP BY animal;
```

Dogs and cats have 2 each. Everyone else has 1.

RUN

WHAT THE COMPUTER SHOWS

animal	how_many
cat	2
dog	2
fish	1
hamster	1
parrot	1
rabbit	1

6 rows came back!

Treats per Animal



YOUR SQL SPELL

```
SELECT animal, SUM(treats) AS treats
FROM pets
GROUP BY animal;
```



The dogs ate 27 treats together! What a hungry bucket.

RUN

WHAT THE COMPUTER SHOWS

animal	treats
cat	17
dog	27
fish	2
hamster	3
parrot	4
rabbit	9

6 rows came back!

Owners per Town



YOUR SQL SPELL

```
SELECT town, COUNT(*) AS people
FROM owners
GROUP BY town;
```

RUN

WHAT THE COMPUTER SHOWS

town	people
Moonbridge	2
Starfield	1
Sunnyvale	2

3 rows came back!

Moonbridge and Sunnyvale have 2 owners each.

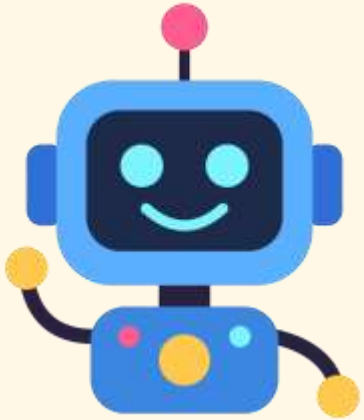
YOUR TURN!

Colour Buckets



Count how many pets there are of each colour.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT colour, COUNT(*)
FROM pets
_____ __ colour;
```

Write your answer here:

Hint: GROUP BY the column.

ANSWER CHECK

Answer: Colour Buckets



YOUR SQL SPELL

```
SELECT colour, COUNT(*) AS pets
FROM pets
GROUP BY colour;
```

Orange and white have 2 pets each.

RUN

WHAT THE COMPUTER SHOWS

colour	pets
black	1
brown	1
gold	1
green	1
orange	2
white	2

6 rows came back!

RECAP

Super Sticker Recap: GROUP BY



- GROUP BY makes buckets
- COUNT and SUM work on each bucket
- Each bucket gives back one row



Bucket Boss Badge Earned!

22

MODULE 22 OF 35

HAVING

A filter just for buckets

Mission: Keep only the groups you want



MODULE 22 · HAVING

WHERE for Groups



HAVING is like WHERE, but it checks whole buckets after GROUP BY.



WHERE

Checks single rows, before grouping.



HAVING

Checks buckets, after grouping.



The Order

GROUP BY comes first, then HAVING.

WHERE or HAVING?



WHERE

Looks at each row,
like one single pet.

HAVING

Looks at each bucket,
like all the dogs together.

If you are checking COUNT or SUM, you need HAVING!

Buckets With 2 or More



YOUR SQL SPELL

```
SELECT animal, COUNT(*) AS pets
FROM pets
GROUP BY animal
HAVING COUNT(*) > 1;
```

RUN

WHAT THE COMPUTER SHOWS

animal	pets
cat	2
dog	2

2 rows came back!

Only the dog and cat buckets have more than 1 pet.

MODULE 22 · HAVING

Hungry Buckets



YOUR SQL SPELL

```
SELECT animal, SUM(treats) AS treats
FROM pets
GROUP BY animal
HAVING SUM(treats) >= 10;
```

🔍 These buckets ate 10 treats or more.

RUN

WHAT THE COMPUTER SHOWS

animal	treats
cat	17
dog	27

2 rows came back!

YOUR TURN!

Busy Towns



Show the towns that have 2 or more owners.

You can do it! Try first, then peek at the next page.

FILL IN THE BLANKS

```
SELECT town, COUNT(*)
FROM owners
GROUP BY town
_____ COUNT(*) >= 2;
```



Hint: Filter the buckets.

ANSWER CHECK

Answer: Busy Towns



YOUR SQL SPELL

```
SELECT town, COUNT(*) AS people
FROM owners
GROUP BY town
HAVING COUNT(*) >= 2;
```

RUN

WHAT THE COMPUTER SHOWS

town	people
Moonbridge	2
Sunnyvale	2

2 rows came back!

Moonbridge and Sunnyvale are the busy towns.

RECAP

Super Sticker Recap: HAVING



- HAVING filters buckets
- WHERE filters single rows
- HAVING comes after GROUP BY



Bucket Filter Badge Earned!

23

MODULE 23 OF 35

NULL: The Empty Box

When we just do not know

Mission: Find and handle missing data



What Is NULL?



NULL means “no value here”. It is an empty box, not zero and not a word!



Empty Box

Nothing has been put inside.



Not Zero

0 is a number. NULL is nothing at all.



Bubbles

Bubbles the fish has no owner, so owner_id is NULL.

Who Has No Owner?



YOUR SQL SPELL

```
SELECT name, owner_id
FROM pets
WHERE owner_id IS NULL;
```

RUN

WHAT THE COMPUTER SHOWS

name	owner_id
Bubbles	NULL

1 row came back!

Bubbles lives in the classroom tank, so there is no owner!

MODULE 23 · NULL

Pets With Owners



YOUR SQL SPELL

```
SELECT name, owner_id
FROM pets
WHERE owner_id IS NOT NULL;
```

IS NOT NULL keeps only the boxes that have something inside.

RUN

WHAT THE COMPUTER SHOWS

name	owner_id
Biscuit	1
Luna	2
Pip	3
Rex	1
Mango	4
Coco	2
Ziggy	3

7 rows came back!

MODULE 23 · NULL

Careful With NULL





Oops!

```sql
WHERE owner_id = NULL
```



Hooray!

```sql
WHERE owner_id IS NULL
```



NULL is special. Always use IS NULL, never = NULL.

MODULE 23 · NULL

Fill the Empty Box



YOUR SQL SPELL

```
SELECT name,
COALESCE(owner_id, 0) AS owner
FROM pets;
```

COALESCE swaps an empty box for something else. Here NULL becomes 0.

RUN

WHAT THE COMPUTER SHOWS

name	owner
Biscuit	1
Luna	2
Pip	3
Rex	1
Mango	4
Coco	2
Ziggy	3
Bubbles	0

8 rows came back!

YOUR TURN!

Count the Empty Boxes



Count how many pets have NO owner.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT COUNT(*)
FROM pets
WHERE owner_id __ ____;
```

Write your answer here:

Hint: Use IS NULL.

ANSWER CHECK

Answer: Empty Boxes



YOUR SQL SPELL

```
SELECT COUNT(*) AS no_owner
FROM pets
WHERE owner_id IS NULL;
```

RUN

WHAT THE COMPUTER SHOWS

no_owner
1

1 row came back!

Just 1 pet: Bubbles the fish.

RECAP

Super Sticker Recap: NULL: The Empty Box



- NULL means no value
- Use IS NULL and IS NOT NULL
- COALESCE fills empty boxes



Empty Box Expert Badge Earned!

24

MODULE 24 OF 35

CREATE TABLE

Build your very own table

Mission: Make a brand new empty table



MODULE 24 · CREATE

Build a New Shelf



CREATE TABLE makes a brand new, empty table. You choose its name and its columns!



Name It

Give the table a name, like toys.



Columns

List each column with its type.



Empty at First

It starts with zero rows.

MODULE 24 · CREATE

The Builder Spell



Here is the magic spell. Every piece has its own job!

`CREATE TABLE`	`toys`	`(`	`id INTEGER,`	`toy TEXT`	`);`
build a table	**called toys**	**open the list**	**a number column**	**a words column**	**close and finish**



Brackets hold the list of columns. Commas go between the columns.

MODULE 24 · CREATE

Make a Toys Table



YOUR SQL SPELL

```
CREATE TABLE toys (
    id INTEGER PRIMARY KEY,
    toy TEXT,
    colour TEXT,
    coins INTEGER
);
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins

0 rows: nothing here yet!



The table exists now, with 4 columns but no rows yet. Like an empty shelf!

BIG IDEA

A new table is like a brand new notebook.

Lots of space, and nothing written in it yet!



YOUR TURN!

Snack Table



Create a table called snacks with columns id (INTEGER), snack (TEXT) and yummy (INTEGER).

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
CREATE TABLE _____ (
    id _______,
    snack ____,
    yummy _______
);
```

Hint: Number columns are INTEGER. Word columns are TEXT.

ANSWER CHECK

Answer: Snack Table



YOUR SQL SPELL

```
CREATE TABLE snacks (
    id INTEGER,
    snack TEXT,
    yummy INTEGER
);
SELECT * FROM snacks;
```

RUN

WHAT THE COMPUTER SHOWS

id	snack	yummy

0 rows: nothing here yet!

Your snacks table is ready and waiting for snacks!

RECAP

Super Sticker Recap: CREATE TABLE



- CREATE TABLE makes a new table
- List the columns inside brackets
- A new table starts empty



Table Builder Badge Earned!

25

MODULE 25 OF 35

Data Types

Every column has a kind

Mission: Choose the right type for each column



MODULE 25 · TYPES

Kinds of Data



Each column holds one kind of data. That kind is called its TYPE.



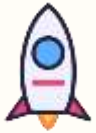
INTEGER

Whole numbers: 1, 7, 100



TEXT

Words in quotes: 'Luna'



REAL

Numbers with a dot: 2.5

Ask the Computer



YOUR SQL SPELL

```
SELECT TYPEOF(7) AS a,
TYPEOF('Luna') AS b,
TYPEOF(2.5) AS c;
```

RUN

WHAT THE COMPUTER SHOWS

a	b	c
integer	text	real

1 row came back!

TYPEOF tells us the type of each value. Clever!

Yes or No Data



For yes or no facts, we often store 1 for yes and 0 for no.



1 Means Yes

Is it fluffy? 1!



0 Means No

Can it fly? 0!



Use INTEGER

1 and 0 are whole numbers.

QUIZ · MODULE 25

Quick Quiz Time!



Which type fits a height like 1.25?

A TEXT

B INTEGER

C REAL



Choose A, B or C. Then turn the page upside down to check the yellow box!

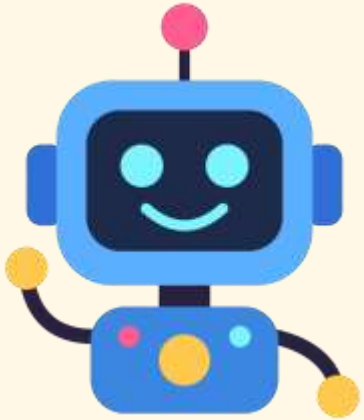
Answer: C. It has a dot, so it is REAL.

YOUR TURN!

Match the Types

Choose a type for each column: a pet’s name, a pet’s age, and a pet’s weight like 3.5 kg.

You can do it! Try first, then peek at the next page.



Write your answer here:

Hint: Words, whole numbers, numbers with a dot.

ANSWER CHECK

Answer: Match the Types



YOUR SQL SPELL

```
CREATE TABLE weights (
    name TEXT,
    age INTEGER,
    weight REAL
);
SELECT * FROM weights;
```

RUN

WHAT THE COMPUTER SHOWS

name	age	weight

0 rows: nothing here yet!

name is TEXT, age is INTEGER and weight is REAL.

RECAP

Super Sticker Recap: Data Types



- INTEGER means whole numbers
- TEXT means words
- REAL means numbers with a dot



Type Teller Badge Earned!

26

MODULE 26 OF 35

INSERT INTO

Put new rows into a table

Mission: Add new data to a table



MODULE 26 · INSERT

Filling the Shelf



INSERT INTO adds new rows. It is like putting new toys on the shelf!



INTO

Say which table.



VALUES

List the facts in brackets.



Right Order

Values go in the same order as the columns.

MODULE 26 · INSERT

The Adding Spell



Here is the magic spell. Every piece has its own job!

```
INSERT INTO  toys  VALUES  (4, 'Teddy', 'brown', 10)  ;
```

- **INSERT INTO** — put into
- **toys** — the toys table
- **VALUES** — these facts
- **(4, 'Teddy', 'brown', 10)** — one new row
- **;** — the end



Our toys table already has a Bouncy Ball, a Kite and a Robot. Let’s add more!

MODULE 26 · INSERT

Add a Teddy



YOUR SQL SPELL

```
INSERT INTO toys
VALUES (4, 'Teddy', 'brown', 10);
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	blue	8
3	Robot	silver	15
4	Teddy	brown	10

4 rows came back!

Teddy joined the shelf as toy number 4!

MODULE 26 · INSERT

Add Two at Once



YOUR SQL SPELL

```
INSERT INTO toys VALUES
(4, 'Yo Yo', 'green', 2),
(5, 'Drum', 'red', 6);
SELECT * FROM toys;
```



Two new toys in one spell. Put a comma between the rows.

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	blue	8
3	Robot	silver	15
4	Yo Yo	green	2
5	Drum	red	6

5 rows came back!

Name the Columns



YOUR SQL SPELL

```
INSERT INTO toys (id, toy)
VALUES (4, 'Puzzle');
SELECT * FROM toys;
```

RUN

We only gave id and toy. The other boxes are NULL, empty!

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	blue	8
3	Robot	silver	15
4	Puzzle	NULL	NULL

4 rows came back!

YOUR TURN!

Add a Rocket

Add a toy: id 4, a Rocket, colour white, costing 12 coins.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
INSERT INTO ____
VALUES (_, '_____', '_____', __);
```

Write your answer here:

Hint: Words go in quotes, numbers do not.

ANSWER CHECK

Answer: Add a Rocket



YOUR SQL SPELL

```
INSERT INTO toys
VALUES (4, 'Rocket', 'white', 12);
SELECT * FROM toys;
```

The Rocket is on the shelf, ready for blast off!

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	blue	8
3	Robot	silver	15
4	Rocket	white	12

4 rows came back!

RECAP

Super Sticker Recap: INSERT INTO



- INSERT INTO adds new rows
- VALUES lists the facts in order
- You can add many rows at once



Shelf Filler Badge Earned!

27

MODULE 27 OF 35

UPDATE

Change facts that are already there

Mission: Fix or change existing data



MODULE 27 · UPDATE

Changing Facts



UPDATE changes data that is already in a table. Like fixing a spelling in your notebook!



SET

Gives the new value.



WHERE

Says WHICH rows to change.



Careful!

No WHERE means EVERY row changes!

MODULE 27 · UPDATE

The Changing Spell



Here is the magic spell. Every piece has its own job!

```
UPDATE toys
SET coins = 5
WHERE id = 1
;
```

change the toys table

make coins 5

only for toy 1

the end



SET says what changes. WHERE says which row changes.

Price Change



YOUR SQL SPELL

```
UPDATE toys
SET coins = 5
WHERE id = 1;
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	5
2	Kite	blue	8
3	Robot	silver	15

3 rows came back!

The Bouncy Ball now costs 5 coins instead of 3.

MODULE 27 · UPDATE

Change Two Things



YOUR SQL SPELL

```
UPDATE toys
SET colour = 'gold', coins = 20
WHERE toy = 'Robot';
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	blue	8
3	Robot	gold	20

3 rows came back!

The Robot is now gold and costs 20 coins.

Always Use WHERE



Oops!

```sql
UPDATE toys
SET coins = 0;
```

Hooray!

```sql
UPDATE toys
SET coins = 0
WHERE id = 2;
```

Without WHERE, EVERY single toy would cost 0 coins!

YOUR TURN!

Blue to Pink





Change the Kite's colour to pink.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
UPDATE toys
___ colour = '____'
WHERE toy = '____';
```

Write your answer here:

Hint: SET gives the new value.

ANSWER CHECK

Answer: Blue to Pink



YOUR SQL SPELL

```
UPDATE toys
SET colour = 'pink'
WHERE toy = 'Kite';
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	pink	8
3	Robot	silver	15

3 rows came back!

The Kite is pink now. So pretty!

RECAP

Super Sticker Recap: UPDATE



- UPDATE changes rows that already exist
- SET gives the new value
- Always add a WHERE



Fact Fixer Badge Earned!

28

MODULE 28 OF 35

DELETE

Remove rows you do not need

Mission: Delete rows safely



MODULE 28 · DELETE

Tidy Up Time



DELETE removes rows from a table. Like taking a broken toy off the shelf.



DELETE FROM

Say which table.



WHERE

Say WHICH rows should go.



No Undo!

Deleted rows are gone for good.

MODULE 28 · DELETE

Goodbye, Kite



YOUR SQL SPELL

```
DELETE FROM toys
WHERE toy = 'Kite';
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
3	Robot	silver	15

2 rows came back!

The Kite flew away. Only 2 toys are left.

MODULE 28 · DELETE

Remove Cheap Toys



YOUR SQL SPELL

```
DELETE FROM toys
WHERE coins < 10;
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
3	Robot	silver	15

1 row came back!

Every toy under 10 coins is gone. Only the Robot stays.

MODULE 28 · DELETE

The Scary Spell



Oops!

SQL

```
DELETE FROM toys;
```

Hooray!

SQL

```
DELETE FROM toys
WHERE id = 3;
```

DELETE without WHERE empties the WHOLE table!

See the Danger



YOUR SQL SPELL

```
DELETE FROM toys;
SELECT * FROM toys;
```

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins

0 rows: nothing here yet!

Uh oh! Zero rows. The whole shelf is empty. Always check your WHERE!

YOUR TURN!

Remove the Robot



Delete the toy with id 3.

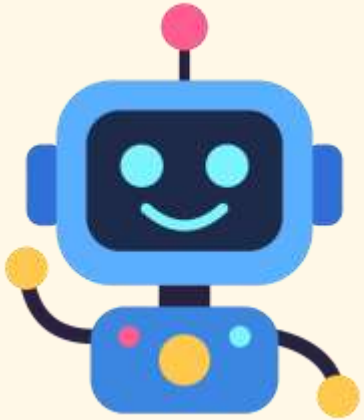
FILL IN THE BLANKS

_____ FROM toys
WHERE __ = _;

Write your answer here:

Hint: DELETE FROM, then WHERE.

You can do it! Try first, then peek at the next page.



ANSWER CHECK

Answer: Remove the Robot



YOUR SQL SPELL

```
DELETE FROM toys
WHERE id = 3;
SELECT * FROM toys;
```

The Robot has gone. 2 toys are left.

RUN

WHAT THE COMPUTER SHOWS

id	toy	colour	coins
1	Bouncy Ball	red	3
2	Kite	blue	8

2 rows came back!

RECAP

Super Sticker Recap: DELETE



- DELETE removes rows
- Always use a WHERE
- There is no undo button



Safe Deleter Badge Earned!

29

MODULE 29 OF 35

Keys: Name Tags for Rows

How tables point to each other

Mission: Learn primary keys and foreign keys



MODULE 29 · KEYS

The Primary Key



A primary key is a special column that is different for EVERY row, like a name tag.



Unique

No two rows ever share it.



Never Empty

It must always have a value.



Our Keys

id is the key in pets and in owners.

MODULE 29 · KEYS

The Foreign Key



A foreign key points to a row in ANOTHER table. It is an arrow saying “look over there!”



Points Away

pets.owner_id points to owners.id



Connects

It tells us who owns which pet.



Example

Pip has owner_id 3, so Priya owns Pip!

MODULE 29 · KEYS

Follow the Matching Numbers



OWNERS TABLE

id	name	town
1	Amara	Sunnyvale
2	Leo	Moonbridge
3	Priya	Sunnyvale
4	Kofi	Starfield
5	Mia	Moonbridge



Same number, same owner! Biscuit and Rex have owner_id 1, so they both belong to Amara.

PETS TABLE

name	owner_id
Biscuit	1
Luna	2
Pip	3
Rex	1
Mango	4
Coco	2
Ziggy	3
Bubbles	NULL

Pets and Owner Numbers



YOUR SQL SPELL

```
SELECT name, owner_id
FROM pets;
```

Each owner_id is a pointer to a row in the owners table.

RUN

WHAT THE COMPUTER SHOWS

name	owner_id
Biscuit	1
Luna	2
Pip	3
Rex	1
Mango	4
Coco	2
Ziggy	3
Bubbles	NULL

8 rows came back!

QUIZ · MODULE 29

Quick Quiz Time!



Rex has owner_id 1. Who owns Rex?

A Leo

B Amara

C Kofi



Choose A, B or C. Then turn the page upside down to check the yellow box!

Answer: B. Owner number 1 is Amara.

YOUR TURN!

Who Owns Mango?

Mango has owner_id 4. Write a spell to find owner number 4.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name
FROM owners
WHERE __ = _;
```

Write your answer here:

Hint: Look in the owners table for id 4.

ANSWER CHECK

Answer: Mango’s Owner



YOUR SQL SPELL

```
SELECT name, town
FROM owners
WHERE id = 4;
```

RUN

WHAT THE COMPUTER SHOWS

name	town
Kofi	Starfield

1 row came back!

Kofi from Starfield owns Mango the parrot!

RECAP

Super Sticker Recap: Keys: Name Tags for Rows



- A primary key is unique for every row
- A foreign key points to another table
- Matching numbers connect the tables



Key Keeper Badge Earned!

30

MODULE 30 OF 35

INNER JOIN

Glue two tables together

Mission: Combine pets with their owners



MODULE 30 · JOIN

Why Join?



Pet names live in one table. Owner names live in another. JOIN glues them together!

Glue

Matches rows from two tables.

ON

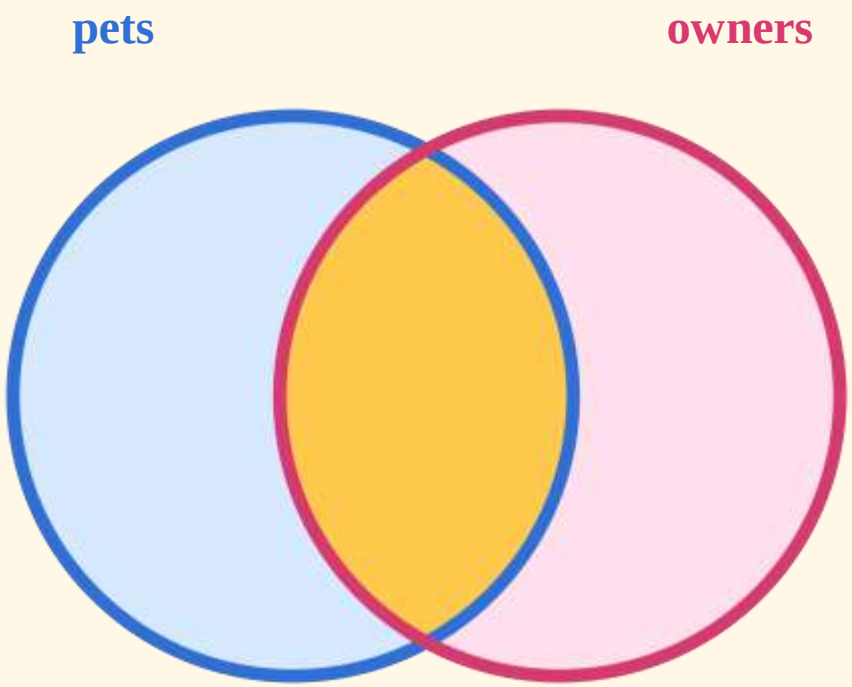
Says which columns must match.

Matches Only

INNER JOIN keeps only rows that match.

The Middle Part





Yellow part = the rows that come back



INNER JOIN keeps only the yellow middle.



Bubbles has no owner, so the fish is left out.



Mia has no pet, so she is left out too.

The Glue Spell



Here is the magic spell. Every piece has its own job!

```
SELECT ...
```
the columns

```
FROM pets
```
start with pets

```
JOIN owners
```
glue on owners

```
ON pets.owner_id = owners.id
```
where the numbers match

```
;
```
the end



JOIN on its own means INNER JOIN. Both words work!

MODULE 30 · JOIN

Pets With Owners



YOUR SQL SPELL

```
SELECT pets.name AS pet,
owners.name AS owner
FROM pets
JOIN owners
ON pets.owner_id = owners.id;
```

RUN

7 pets found their owners. Bubbles has no owner, so the fish is left out.

WHAT THE COMPUTER SHOWS

pet	owner
Biscuit	Amara
Luna	Leo
Pip	Priya
Rex	Amara
Mango	Kofi
Coco	Leo
Ziggy	Priya

7 rows came back!

Sunnyvale Pets



YOUR SQL SPELL

```
SELECT pets.name AS pet,
owners.town
FROM pets
JOIN owners
ON pets.owner_id = owners.id
WHERE owners.town = 'Sunnyvale';
```

RUN

WHAT THE COMPUTER SHOWS

pet	town
Biscuit	Sunnyvale
Pip	Sunnyvale
Rex	Sunnyvale
Ziggy	Sunnyvale

4 rows came back!

These pets have owners who live in Sunnyvale.

MODULE 30 · JOIN

Short Table Names



YOUR SQL SPELL

```
SELECT p.name AS pet, o.name AS owner
FROM pets AS p
JOIN owners AS o
ON p.owner_id = o.id
LIMIT 4;
```

p and o are short nicknames for the tables. Less typing!

RUN

WHAT THE COMPUTER SHOWS

pet	owner
Biscuit	Amara
Luna	Leo
Pip	Priya
Rex	Amara

4 rows came back!

YOUR TURN!

Leo’s Pets

Show the names of the pets that belong to Leo.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT p.name
FROM pets AS p
JOIN owners AS o
ON p.owner_id = o.id
WHERE o.name = '___';
```

Hint: Join first, then filter by the owner name.

ANSWER CHECK

Answer: Leo's Pets



YOUR SQL SPELL

```
SELECT p.name AS pet
FROM pets AS p
JOIN owners AS o
ON p.owner_id = o.id
WHERE o.name = 'Leo';
```

RUN

WHAT THE COMPUTER SHOWS

pet
Luna
Coco

2 rows came back!

Luna and Coco, the two cats, belong to Leo.

RECAP

Super Sticker Recap: INNER JOIN

- JOIN glues two tables together
- ON says which columns match
- INNER JOIN keeps only the matches





Table Gluer Badge Earned!

31

MODULE 31 OF 35

LEFT JOIN

Keep everyone from the left side

Mission: Join without losing anybody



MODULE 31 · LEFT JOIN

Nobody Left Behind



LEFT JOIN keeps EVERY row from the first table, even with no match. Missing parts become NULL.



Keeps All

Every row from the left table stays.



Empty Boxes

No match? You get NULL.

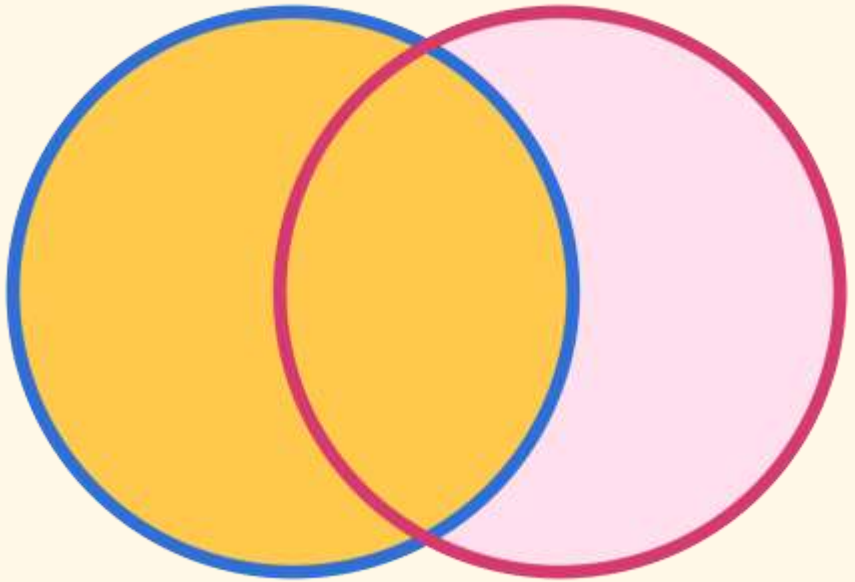


Find the Lonely

Great for finding things with no partner.

The Whole Left Circle





Yellow part = the rows that come back

LEFT JOIN keeps the whole left circle.

Mia has no pet, but she still comes back.

Her pet box will be NULL, empty.

MODULE 31 · LEFT JOIN

Every Owner



YOUR SQL SPELL

```
SELECT o.name AS owner, p.name AS pet
FROM owners AS o
LEFT JOIN pets AS p
ON p.owner_id = o.id;
```

Mia has no pet yet, so her pet box is NULL. But she is still here!

RUN

WHAT THE COMPUTER SHOWS

owner	pet
Amara	Biscuit
Amara	Rex
Leo	Coco
Leo	Luna
Priya	Pip
Priya	Ziggy
Kofi	Mango
Mia	NULL

8 rows came back!

MODULE 31 · LEFT JOIN

Owners With No Pets



YOUR SQL SPELL

```
SELECT o.name AS owner
FROM owners AS o
LEFT JOIN pets AS p
ON p.owner_id = o.id
WHERE p.id IS NULL;
```

RUN

WHAT THE COMPUTER SHOWS

owner
Mia

1 row came back!

LEFT JOIN plus IS NULL finds the lonely ones. Mia needs a pet!

JOIN or LEFT JOIN?



INNER JOIN

Only rows that match.
Mia disappears.

LEFT JOIN

Every left row stays.
Mia stays, with NULL.

Pick LEFT JOIN when you must not lose anybody.

YOUR TURN!

Every Single Pet

Show EVERY pet with its owner's name, even pets with no owner.

You can do it! Try first, then peek at the next page.

FILL IN THE BLANKS

```
SELECT p.name, o.name
FROM pets AS p
____ ____ owners AS o
ON p.owner_id = o.id;
```



Hint: Pets are on the left, so use LEFT JOIN.

ANSWER CHECK

Answer: Every Single Pet



YOUR SQL SPELL

```
SELECT p.name AS pet, o.name AS owner
FROM pets AS p
LEFT JOIN owners AS o
ON p.owner_id = o.id;
```

Bubbles stays, with NULL as the owner!

RUN

WHAT THE COMPUTER SHOWS

pet	owner
Biscuit	Amara
Luna	Leo
Pip	Priya
Rex	Amara
Mango	Kofi
Coco	Leo
Ziggy	Priya
Bubbles	NULL

8 rows came back!

RECAP

Super Sticker Recap: LEFT JOIN



- LEFT JOIN keeps all the left rows
- No match gives NULL
- LEFT JOIN plus IS NULL finds the lonely ones



Left Keeper Badge Earned!

32

MODULE 32 OF 35

Subqueries

A question inside a question

Mission: Use one query inside another



MODULE 32 · SUBQUERY

Boxes Inside Boxes



A subquery is a small query tucked inside a bigger one, like a box inside a box!



Inside First

The inside query runs first.



Then Outside

Its answer is used by the outside query.



Brackets

Put the inside query in brackets.

BIG IDEA

Step 1: find the average. Step 2: find the pets above it!

A subquery lets the computer do both steps in one spell.



MODULE 32 · SUBQUERY

Above Average Eaters



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
WHERE treats > (
    SELECT AVG(treats)
    FROM pets
);
```

RUN

WHAT THE COMPUTER SHOWS

name	treats
Biscuit	12
Pip	9
Rex	15
Coco	10

4 rows came back!

The inside finds the average, 7.75. The outside keeps pets above it.

MODULE 32 · SUBQUERY

Moonbridge Pets



YOUR SQL SPELL

```
SELECT name
FROM pets
WHERE owner_id IN (
    SELECT id FROM owners
    WHERE town = 'Moonbridge'
);
```



The inside finds owners in Moonbridge. The outside finds their pets.

RUN

WHAT THE COMPUTER SHOWS

name
Luna
Coco

2 rows came back!

MODULE 32 · SUBQUERY

The Oldest Pet



YOUR SQL SPELL

```
SELECT name, age
FROM pets
WHERE age = (
    SELECT MAX(age) FROM pets
);
```

RUN

WHAT THE COMPUTER SHOWS

name	age
Rex	7

1 row came back!

No guessing! The subquery finds 7, then we find who is 7.

YOUR TURN!

The Snack King



Find the pet that ate the MOST treats, using a subquery.

You can do it! Try first, then peek at the next page.



FILL IN THE BLANKS

```
SELECT name
FROM pets
WHERE treats = (
    SELECT ___(treats) FROM pets
);
```

Hint: Biggest means MAX.

ANSWER CHECK

Answer: The Snack King



YOUR SQL SPELL

```
SELECT name, treats
FROM pets
WHERE treats = (
    SELECT MAX(treats) FROM pets
);
```

RUN

WHAT THE COMPUTER SHOWS

name	treats
Rex	15

1 row came back!

Rex is the Snack King with 15 treats!

RECAP

Super Sticker Recap: Subqueries



- A subquery is a query inside a query
- The inside runs first
- Put it inside brackets



Box Opener Badge Earned!

33

MODULE 33 OF 35

CASE WHEN

Teach the computer to decide

Mission: Make labels with if this, then that



MODULE 33 · CASE

If This, Then That



CASE lets the computer choose a label for each row, like a traffic light choosing a colour.



WHEN

Checks a rule.



THEN

The label to use if the rule is true.



ELSE

The label if no rule is true.

MODULE 33 · CASE

The Deciding Spell



Here is the magic spell. Every piece has its own job!

```
CASE
  WHEN treats > 10
  THEN 'Hungry'
  ELSE 'Normal'
END
```

start deciding — **if this is true** — **use this label** — **or else this one** — **stop deciding**



Add AS and a nickname after END to name your new column.

MODULE 33 · CASE

Snack Labels



YOUR SQL SPELL

```
SELECT name, treats,
CASE
    WHEN treats > 10 THEN 'Hungry'
    ELSE 'Normal'
END AS appetite
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats	appetite
Biscuit	12	Hungry
Luna	7	Normal
Pip	9	Normal
Rex	15	Hungry
Mango	4	Normal
Coco	10	Normal
Ziggy	3	Normal
Bubbles	2	Normal

8 rows came back!

Biscuit and Rex get the Hungry label.

MODULE 33 · CASE

Baby, Kid or Grown Up



YOUR SQL SPELL

```
SELECT name, age,
CASE
    WHEN age <= 1 THEN 'Baby'
    WHEN age <= 4 THEN 'Kid'
    ELSE 'Grown Up'
END AS stage
FROM pets;
```



The computer checks each WHEN from top to bottom.

RUN

WHAT THE COMPUTER SHOWS

name	age	stage
Biscuit	3	Kid
Luna	5	Grown Up
Pip	1	Baby
Rex	7	Grown Up
Mango	2	Kid
Coco	4	Kid
Ziggy	1	Baby
Bubbles	2	Kid

8 rows came back!

YOUR TURN!

Sunny Labels



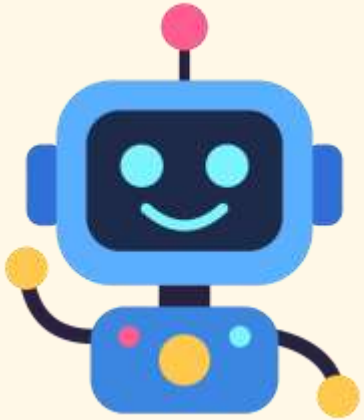
Label each owner Sunny Town if their town is Sunnyvale, and Other Town for everyone else.

FILL IN THE BLANKS

```
SELECT name,
CASE
    WHEN town = '_________' THEN 'Sunny Town'
    ____ 'Other Town'
END AS label
FROM owners;
```

Hint: ELSE catches everyone else.

You can do it! Try first, then peek at the next page.



ANSWER CHECK

Answer: Sunny Labels



YOUR SQL SPELL

```
SELECT name,
CASE
    WHEN town = 'Sunnyvale' THEN 'Sunny Town'
    ELSE 'Other Town'
END AS label
FROM owners;
```

Amara and Priya live in Sunny Town!

RUN

WHAT THE COMPUTER SHOWS

name	label
Amara	Sunny Town
Leo	Other Town
Priya	Sunny Town
Kofi	Other Town
Mia	Other Town

5 rows came back!

RECAP

Super Sticker Recap: CASE WHEN



- CASE chooses a label for each row
- WHEN checks a rule, THEN gives the label
- ELSE catches the rest and END finishes



Decision Maker Badge Earned!

34

MODULE 34 OF 35

Window Functions

Peek at your neighbours

Mission: Rank and number rows without squashing them



MODULE 34 · WINDOWS

Look Through the Window



Window functions look at other rows but keep every row. GROUP BY squashes rows. Windows do not!



OVER

Opens the window and says how to look.



ROW_NUMBER

Gives places 1, 2, 3 in order.



RANK

Like places, but ties share a place.

MODULE 34 · WINDOWS

Snack Race Places



YOUR SQL SPELL

```
SELECT name, treats,
ROW_NUMBER() OVER (
    ORDER BY treats DESC
) AS place
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats	place
Rex	15	1
Biscuit	12	2
Coco	10	3
Pip	9	4
Luna	7	5
Mango	4	6
Ziggy	3	7
Bubbles	2	8

8 rows came back!

Rex is in 1st place in the snack race!

Ties Share a Place



YOUR SQL SPELL

```
SELECT name, age,
RANK() OVER (
    ORDER BY age DESC
) AS age_rank
FROM pets;
```

RUN

WHAT THE COMPUTER SHOWS

name	age	age_rank
Rex	7	1
Luna	5	2
Coco	4	3
Biscuit	3	4
Mango	2	5
Bubbles	2	5
Pip	1	7
Ziggy	1	7

8 rows came back!

Mango and Bubbles are both 2, so they share 5th place. Pip and Ziggy share 7th.

A Race for Each Animal



YOUR SQL SPELL

```
SELECT name, animal, treats,
ROW_NUMBER() OVER (
    PARTITION BY animal
    ORDER BY treats DESC
) AS place
FROM pets;
```



PARTITION BY makes a separate race for each kind of animal.

RUN

WHAT THE COMPUTER SHOWS

name	animal	treats	place
Coco	cat	10	1
Luna	cat	7	2
Rex	dog	15	1
Biscuit	dog	12	2
Bubbles	fish	2	1
Ziggy	hamster	3	1
Mango	parrot	4	1
Pip	rabbit	9	1

8 rows came back!

MODULE 34 · WINDOWS

Running Total



YOUR SQL SPELL

```
SELECT name, treats,
SUM(treats) OVER (
    ORDER BY id
) AS running
FROM pets;
```



The total keeps growing row by row, like filling a jar.

RUN

WHAT THE COMPUTER SHOWS

name	treats	running
Biscuit	12	12
Luna	7	19
Pip	9	28
Rex	15	43
Mango	4	47
Coco	10	57
Ziggy	3	60
Bubbles	2	62

8 rows came back!

YOUR TURN!

The Age Race



Number the pets from youngest to oldest using ROW_NUMBER.

You can do it! Try first, then peek at the next page.

FILL IN THE BLANKS

```
SELECT name, age,
_____() OVER (
  ORDER BY ___
) AS place
FROM pets;
```



Hint: Youngest first means ORDER BY age.

ANSWER CHECK

Answer: The Age Race



YOUR SQL SPELL

```
SELECT name, age,
ROW_NUMBER() OVER (
    ORDER BY age
) AS place
FROM pets;
```

Every pet gets its very own place number.

RUN

WHAT THE COMPUTER SHOWS

name	age	place
Pip	1	1
Ziggy	1	2
Mango	2	3
Bubbles	2	4
Biscuit	3	5
Coco	4	6
Luna	5	7
Rex	7	8

8 rows came back!

RECAP

Super Sticker Recap: Window Functions



- Window functions keep every row
- ROW_NUMBER gives places and RANK shares ties
- PARTITION BY makes groups for the window



Window Watcher Badge Earned!

35

MODULE 35 OF 35

WITH: Helper Queries

Name a query and use it again

Mission: Build big answers from small steps



MODULE 35 · WITH

A Helper Recipe Card



WITH lets you write a helper query first and give it a name. Then you use that name like a table!



Step by Step

Break a big problem into small steps.



Give It a Name

The helper gets its own nickname.



Easy to Read

Your spell becomes tidy and clear.

MODULE 35 · WITH

The Helper Spell



Here is the magic spell. Every piece has its own job!

```
WITH big_eaters AS ( ... )
SELECT * FROM big_eaters
```

WITH — make a helper

big_eaters — called big_eaters

AS (...) — here is its query

SELECT * FROM big_eaters — use it like a table



The helper only lives inside this one spell.

MODULE 35 · WITH

Big Eaters Helper



YOUR SQL SPELL

```
WITH big_eaters AS (
    SELECT name, treats
    FROM pets
    WHERE treats >= 10
)
SELECT * FROM big_eaters
ORDER BY treats DESC;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats
Rex	15
Biscuit	12
Coco	10

3 rows came back!



First we make the helper, then we use it like a table.

MODULE 35 · WITH

Treats per Owner



YOUR SQL SPELL

```
WITH totals AS (
    SELECT owner_id, SUM(treats) AS treats
    FROM pets GROUP BY owner_id
)
SELECT o.name, t.treats
FROM totals AS t
JOIN owners AS o ON t.owner_id = o.id
ORDER BY t.treats DESC;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats
Amara	27
Leo	17
Priya	12
Kofi	4

4 rows came back!



Amara’s pets ate the most: 27 treats!

YOUR TURN!

The Final Boss!



Use WITH to make a helper called cats with only the cats, then show everything in it.

FILL IN THE BLANKS

```
WITH ____ AS (
    SELECT name, treats FROM pets
    WHERE animal = '___'
)
SELECT * FROM ____;
```

Hint: Name the helper, then use that name after FROM.

You can do it! Try first, then peek at the next page.



ANSWER CHECK

Answer: Boss Defeated!



YOUR SQL SPELL

```
WITH cats AS (
    SELECT name, treats FROM pets
    WHERE animal = 'cat'
)
SELECT * FROM cats;
```

RUN

WHAT THE COMPUTER SHOWS

name	treats
Luna	7
Coco	10

2 rows came back!

Boss defeated! Luna and Coco appear. You are amazing!

MODULE 35 · WITH

What Comes Next?

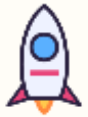


You have reached the start of advanced SQL! Here are some next adventures.



Views

Save a query as a pretend table.



Indexes

Make searches super speedy.



More Joins

Meet RIGHT JOIN and FULL JOIN.



Real Projects

Build a database of your own collection!

RECAP

Super Sticker Recap: WITH: Helper Queries



- WITH makes named helper queries
- Use the helper just like a table
- Big problems become small steps



SQL Superstar Badge Earned!

SQL WORD BANK

Word Bank, Part 1



Word	What it means
SELECT	Show me these columns
FROM	Which table to look in
WHERE	Keep matching rows
*	Every column
AND	Both rules are true
OR	At least one rule
NOT	Flip a rule
IN	Is it on my list?
BETWEEN	Inside a number fence
LIKE	Match a pattern
%	Any letters

Word	What it means
_	Exactly one letter
ORDER BY	Sort the rows
ASC / DESC	Small to big / big to small
LIMIT	Show only a few rows
DISTINCT	Remove repeats
AS	Give a nickname
COUNT	How many rows
SUM	Add them all up
AVG	The fair share
MIN / MAX	Smallest / biggest
ROUND	Tidy a long number

SQL WORD BANK

Word Bank, Part 2



Word	What it means
GROUP BY	Put rows in buckets
HAVING	Filter the buckets
NULL	An empty box
IS NULL	Is the box empty?
COALESCE	Fill an empty box
CREATE TABLE	Build a new table
INTEGER	Whole numbers
TEXT	Words
REAL	Numbers with a dot
INSERT INTO	Add new rows
UPDATE ... SET	Change rows

Word	What it means
DELETE	Remove rows
PRIMARY KEY	A unique name tag
FOREIGN KEY	Points to another table
JOIN	Glue two tables
LEFT JOIN	Keep all left rows
Subquery	A query inside a query
CASE WHEN	If this, then that
OVER	Open a window
ROW_NUMBER / RANK	Places in a race
PARTITION BY	A window per group
WITH	A named helper query







Certificate of SQL Awesomeness

This certificate is proudly given to

_____

for finishing all 35 modules of SQL Quest and becoming an official

SQL Superstar!

_____ _____

Professor Hoot Date





THE END

You did it, SQL Superstar!

Keep asking clever questions. The world is full of data waiting for you to explore!

